Static force Analysis →

If the magnitude of inertia forces are small compared to externally applied load, inertia force can be neglected during the analysis of mechanism. Such an analysis is called as static force analysis.

Static Equilibrium →

A body maintain its Equilibrium state if it is remain in the state of rest or uniform motion.

Example → The Earth rotating around Sun.

$F = ma = 0$, $a = \frac{dv}{dt} = 0$

Lifting Crane.

Condition →

(i) Vector sum of all the forces acting on a body is equal to ZERO.

(ii) Vector Sum of all moments acting on a body is equal to zero.

$\Sigma F = 0$, $\Sigma M = 0$.

Equilibrium of two force Member →

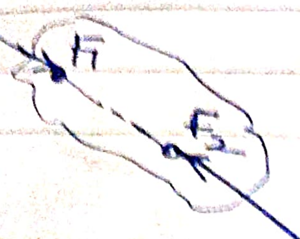
$F_1 = F_2$.

A body under the action of two forces will be equilibrium if

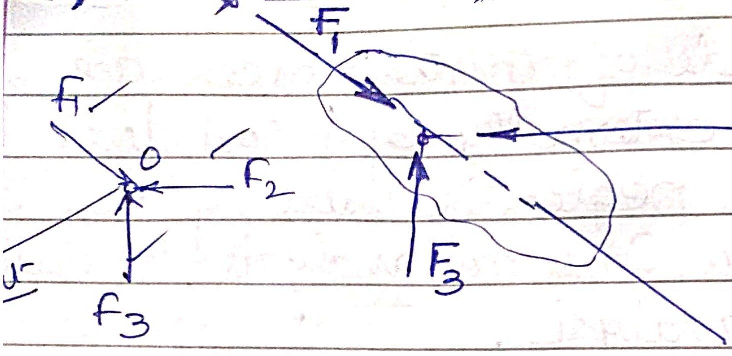
(a) The forces are of same magnitude

(b) Forces act along same line

(c) Forces are in opposite direction.

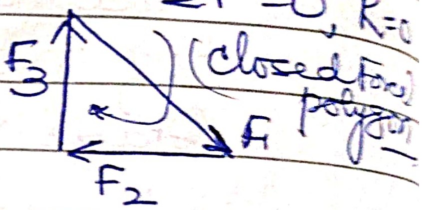


Equilibrium of three force members



$F_1 = F_2 + F_3$

$\Sigma F = 0,\ R = 0$



A body under the action of three forces will be equilibrium if

(a) The resultant force is ZERO.

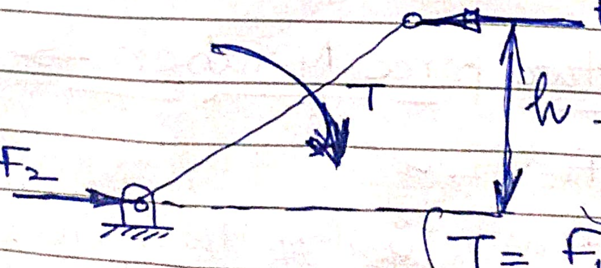
(b) Concurrent forces.

Equilibrium of two forces and Torque

A member under the action of two forces and an applied Torque will be in equilibrium if :

(a) The forces are equal in magnitude, parallel in direction and opposite in nature.

(b) The forces form a couple which is equal and opposite to the applied Torque.



$F_1 = F_2$

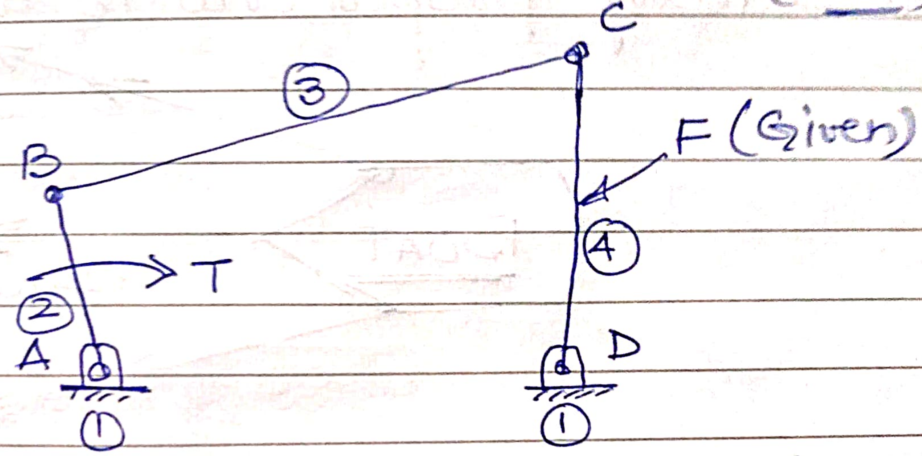
$F_1 \parallel F_2$

$$T = F_1 \times h = F_2 \times h$$

Free Body diagram →

A free body diagram is a diagram of a part isolated from the mechanism in order to determine the nature of forces acting on it

Example → find the force applied on member 2, 3 & 4 (Link)

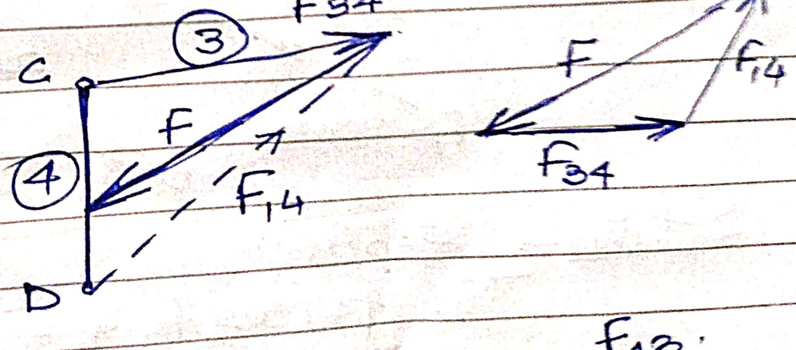


(i) Member (Link) 4 is acted upon by three forces F, F_{34} & F_{14}

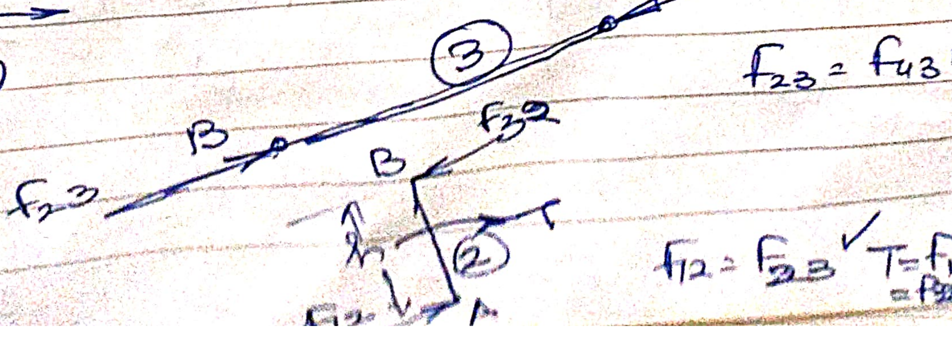
(ii) Member 3 is acted upon by two forces F_{23} and F_{43}

(iii) Member 2 is acted upon by two forces F_{32}, F_{12} and Torque T.

Consider Link 4 →



Consider Link 3 → (2 force member)



$F_{23} = F_{43}$ ✓

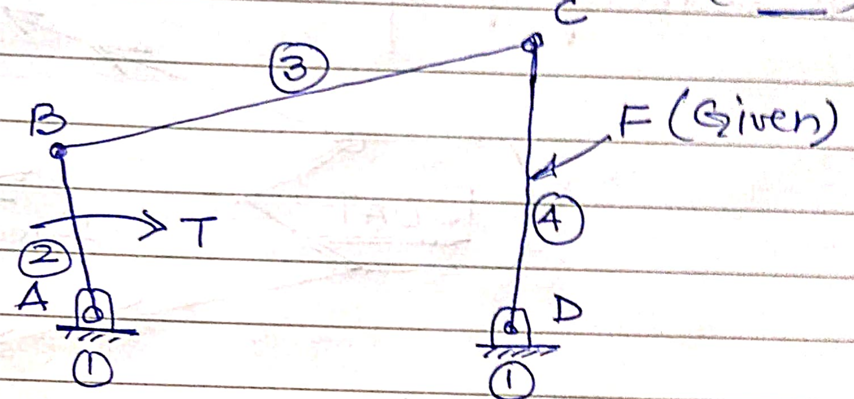
$F_{12} = F_{32}$ ✓ $T = F_{12} \times h$
$= F_{32} \times h$.

$F_{32} = F_{43} = F_{34} = F_{..}$

Free Body diagram →

A free body diagram is a diagram of a part isolated from the mechanism in order to determine the nature of forces acting on it

Example → find the force applied on member 2, 3 & 4. (Link)

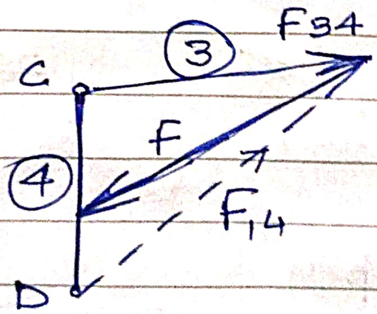


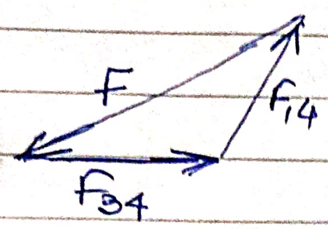
(i) Member (link) 4 is acted upon by three forces F, F_{34} & F_{1}

(ii) Member 3 is acted upon by two forces F_{23} and F_{43}

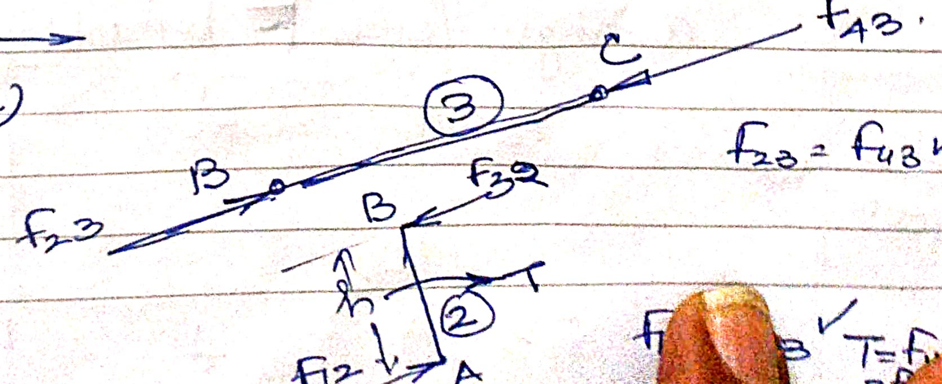
(iii) Member 2 is acted upon by two forces F_{32}, F_{12} and Torque T.

Consider link 4 →





Consider link 3 →
(2 force member)



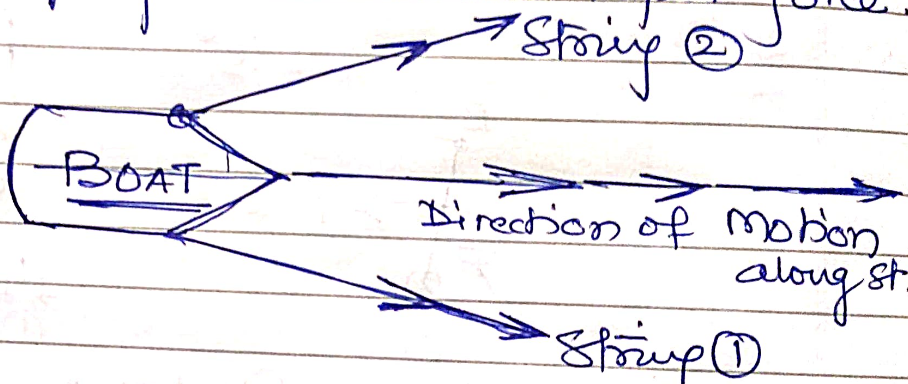
$F_{23} = F_{43}$ ✓

$F_{23} = F_{43} = F_{34} = F_{32}$

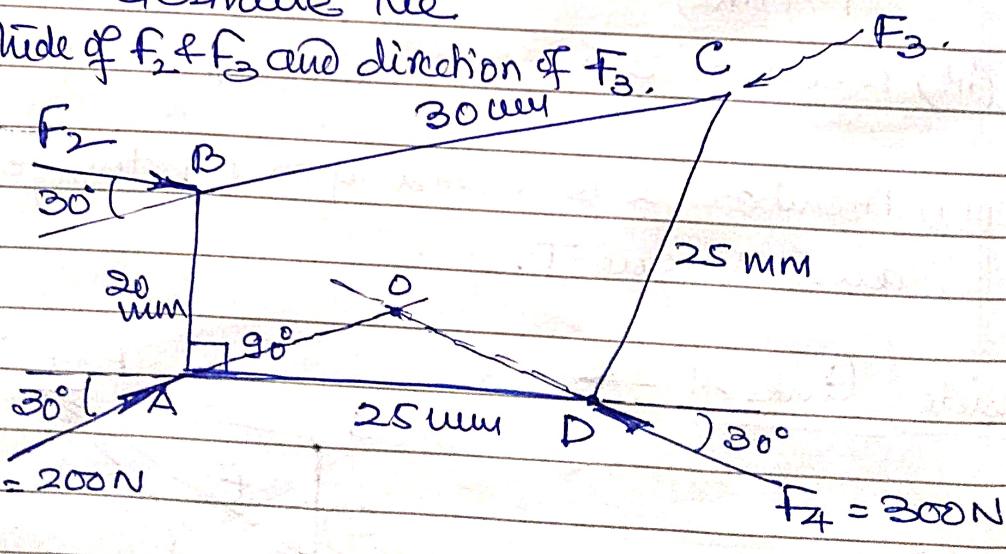
F ... ✓ $T = F \cdot h$

Principle of Superposition →

In linear Systems, if a number of loads act on a System of forces, the net effect is equal to the Superposition of the effects of the individual loads taken one at a time.
Linear System is one in which the output force is directly proportional to the input force.

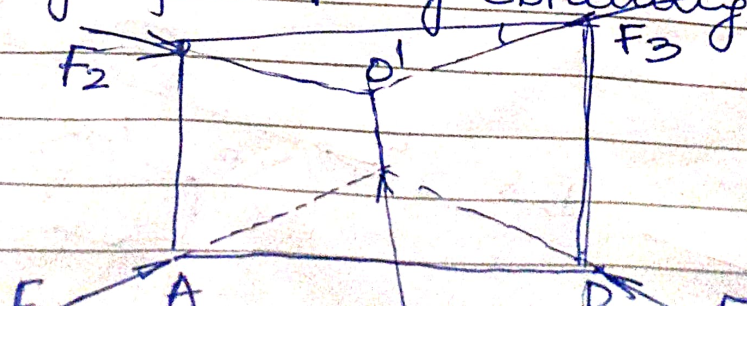


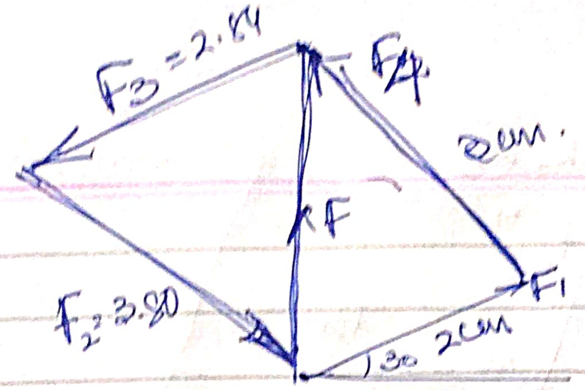
Example → Determine the Magnitude of F_2 & F_3 and direction of F_3.



Solution → Step (i)
The forces F_1 & F_4 can be combined into a single force **F** by obtaining their resultant

Scale
100N = 1cm.



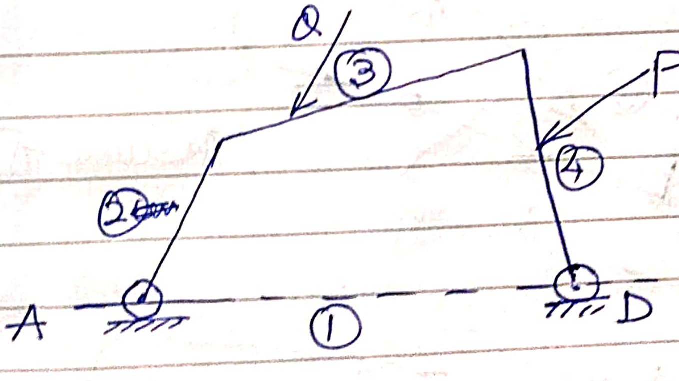


$f_2 = 380$ N

$f_3 = 284$ N.

Prob → Static force analysis of four bar mechanism with 2 known forces. (P & Q).

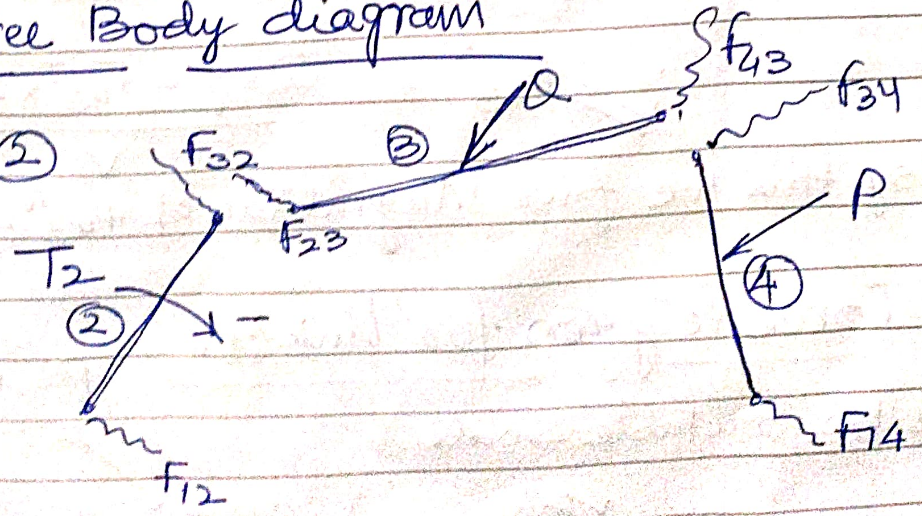
Step ① → Draw Configuration diagram with a Suitable Scale.



Step ②

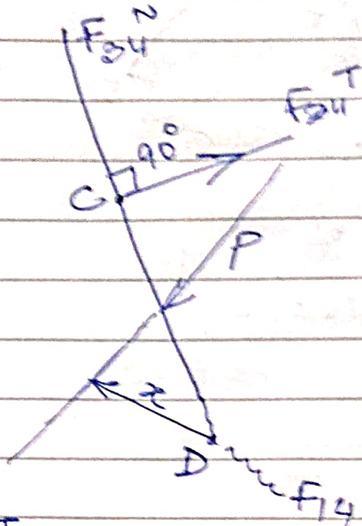
Free Body diagram

at link ②



Step ③ Consider link ④

(Assume the direction of F_{34}^T)



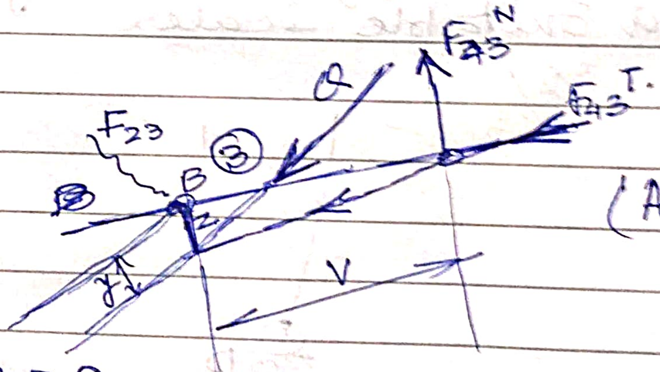
$\Sigma M_D = 0$

Taking Moment about point D.

$P.x = F_{34}^T \times CD.$

$\therefore F_{34}^T = \dfrac{P.x}{CD}$ we find magnitude of F_{34}^T.

Step ④



(Assume F_{43}^N Anticlock wise to B)

$\Sigma M_B = 0$

$$Q \times y + F_{43}^T \times z = F_{43}^N \times V$$

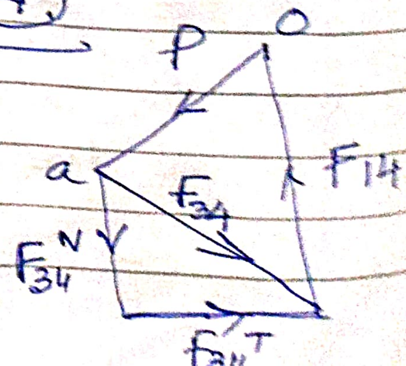
$$\therefore F_{43}^N = \frac{Q \times y + F_{43}^T \times z}{V}.$$

By this we find Magnitude of F_{43}^N & their dir

Step ⑤ Force Polygon for Link ④

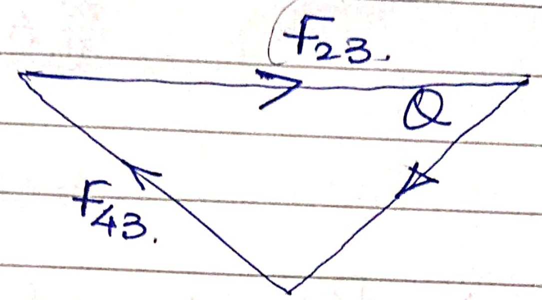
As per Suitable Scale

find F_{34}



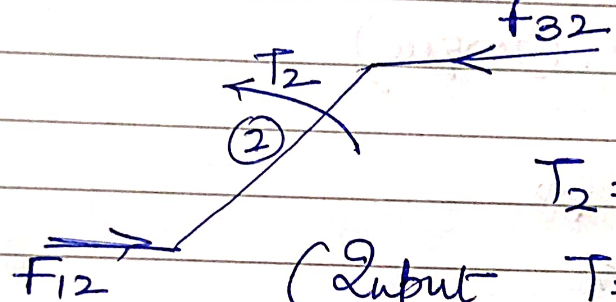
Step (6) force polygon for link (3)

As per Suitable Scale



find F_{23} & its dir^n.

Step (7) for link (2)

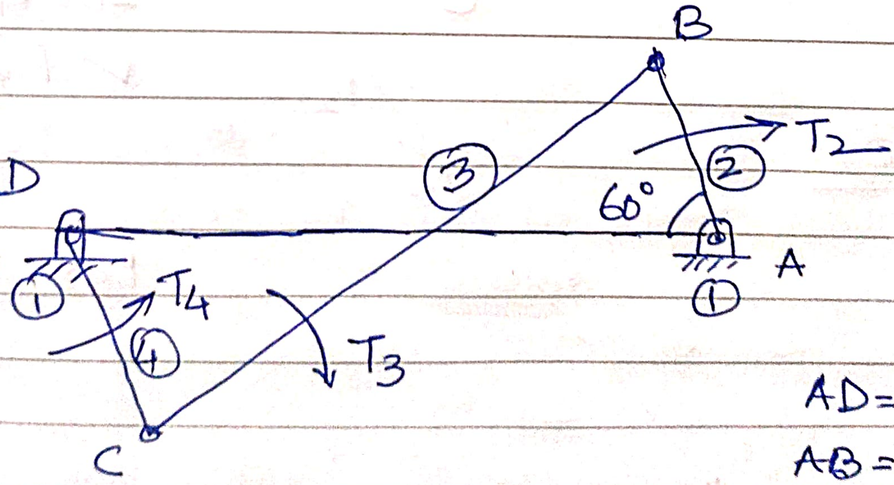


$T_2 = -F_{32} \cdot h.$

(Input Torque) $T = -T_2.$

ie always opposite to couple.

Q.



Given.

$T_3 = 30$ N-m

$T_4 = 20$ N-m.

AD = 800 mm.

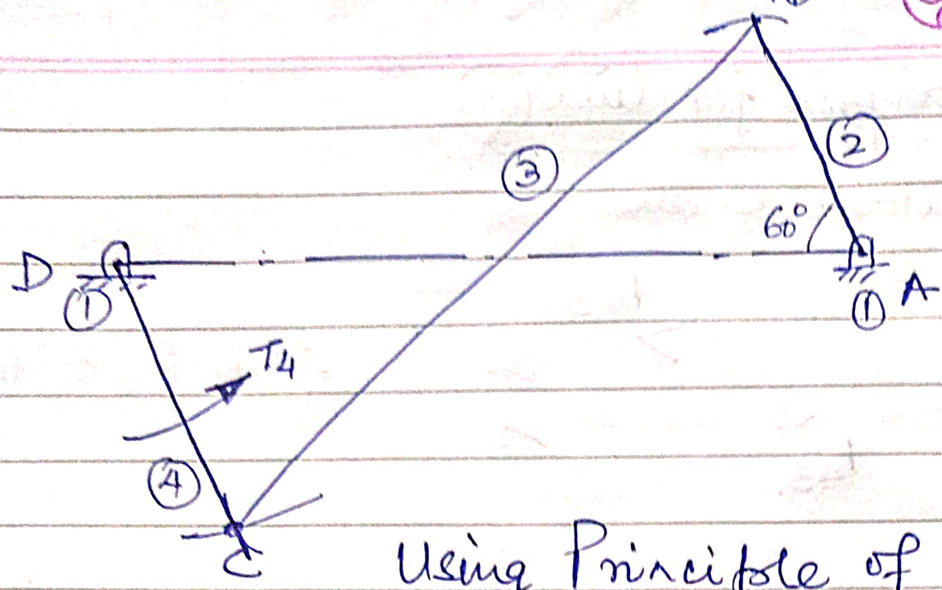
AB = 300 mm.

BC = 700 mm.

CD = 400 mm.

Find $T_2 = ?$

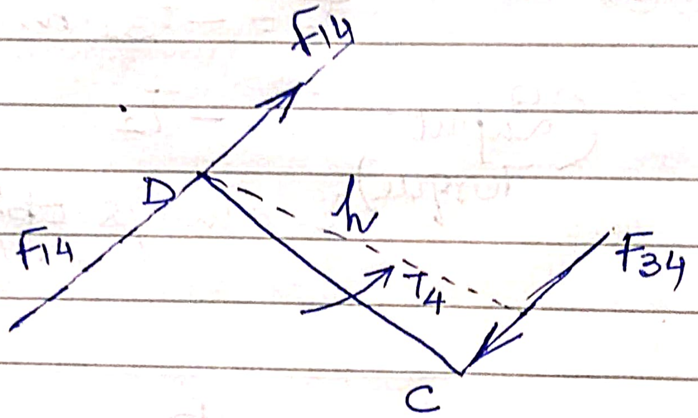
(i) First we draw the Configuration diagram as per Suitable Scale.



Using Principle of Superposition.

Considering T_4. (Torque)

Link 4



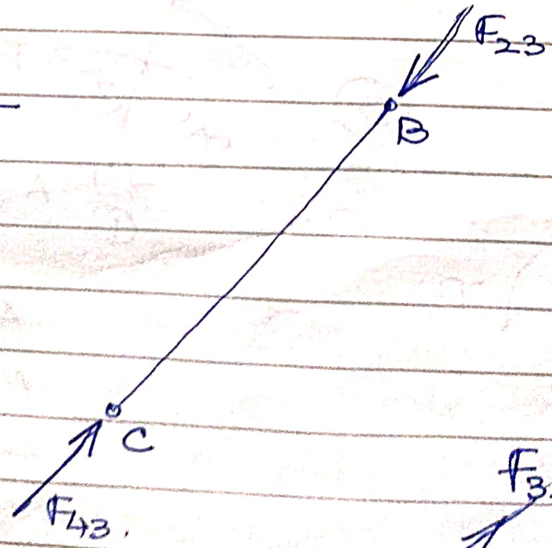
as $F_{34} = F_{14}$

$T_4 = F_{14} \times h$.

$20 = F_{14} \times 3.9 \times \frac{100}{1000}$

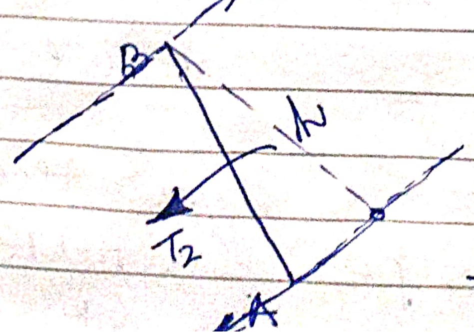
$F_{14} = 51.29\,N$.

Link-3



$F_{43} = F_{23} = F_{32}$

Link 2

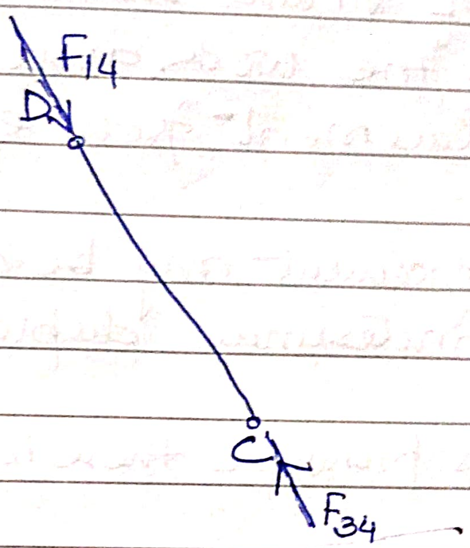


$T_2 = F_{32} \times h$

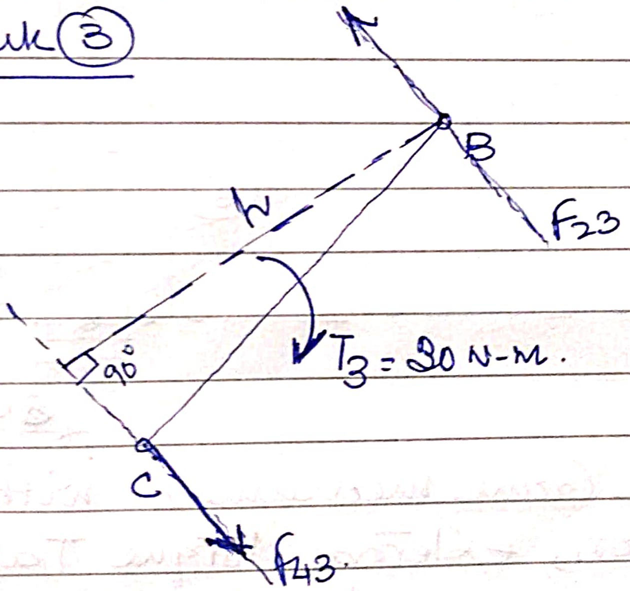
$T_2 = 51.29 \times 2.8 \times \frac{100}{1000}$

Considering Torque (T_3)

link (4)



Link (3)

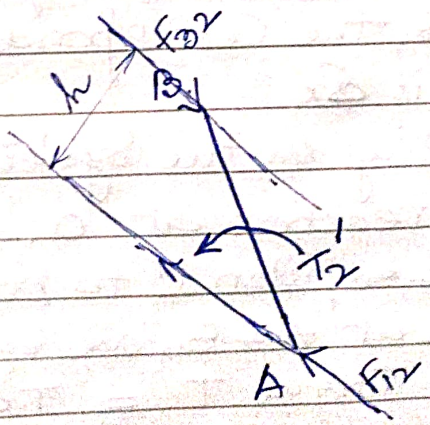


$T_3 = F_{43} \times h$

$30 = F_{43} \times 7 \times \frac{100}{1000}$

$F_{43} = 42.85$ N.

Link (2)



$F_{32} \times h = T_2'$

$42.85 \times 0.5 \times \frac{100}{1000}$

$T_2' = 2.14$ N-m. (CCW)

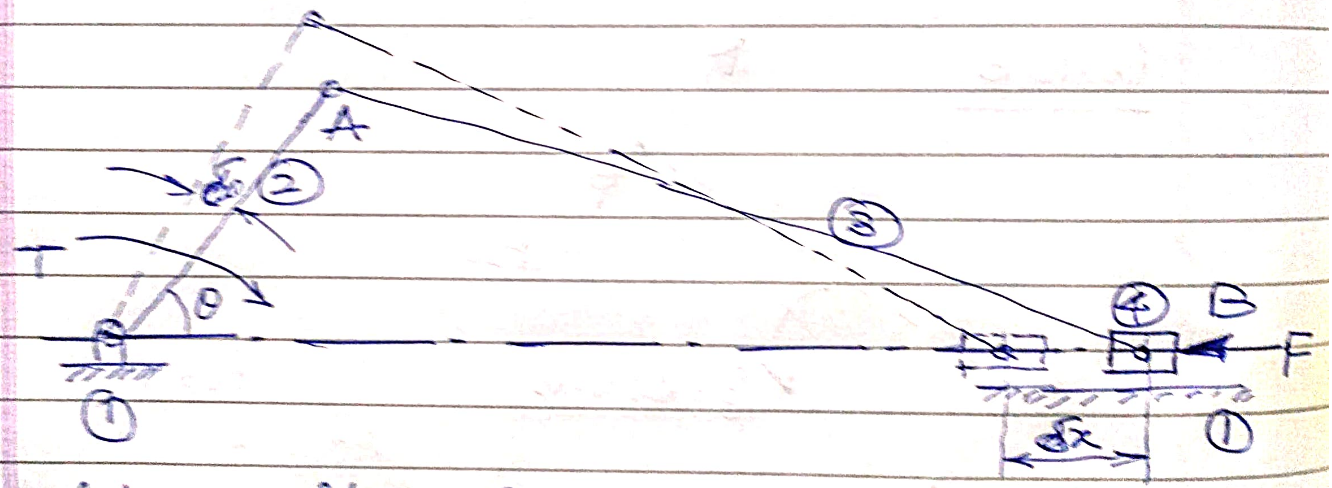
Total Torque at link (2) $= T_2 + T_2'$ (as per dir^n)

Principle of Virtual work

The principle of virtual work (Imaginary) can be stated as "the work done during a virtual displacement" from equilibrium is equal to ZERO.

Virtual displacement may be defined as an imaginary infinitesimal displacement of the system.

In this principle there is no need of FBD.



Consider a Slider Crank mechanism with external force P at piston, external torque T at crank-shaft and force at the bearings.

Let the crank rotates through a small angular displacement $\delta\theta$, the corresponding piston displacement is δx.

The forces acting on the system are

(a) Bearing reaction at O. (No work)

(b) force of cylinder on piston $\perp^r$ to piston displacement (No work)

(c) Bearing force at A & B, are equal & opposite (No work)

(d) work done by Torque T $= T.\delta\theta$

(e) " " " force F $= F.\delta x$

By principle of Virtual work

$$W = T\delta\theta + f\delta x = 0$$

Virtual displacement in time interval δt

$$\cancel{\frac{T\delta\theta}{\delta t}} \quad T.\frac{d\theta}{dt} + f\frac{dx}{dt} = 0$$

$$T.\omega + F v = 0.$$

$$\therefore \boxed{T = -\frac{F}{\omega}.v}$$

$\omega \rightarrow$ Angular velocity

$v \rightarrow$ Linear velocity of piston.

-ve sign indicates that T must be applied in opposite dir^n to the angular velocity for equilibrium.